Question 1:
What is the expected output of the following code?

```
def func(text, num):
    while num > 0:
        print(text)
    num = num - 1

func('Hello', 3)
```

A.	1 | Hello 2 | Hello 3 | Hello 4 | Hello
B.	1 | Hello 2 | Hello
C.	An infinite loop.
D.	1 | Hello 2 | Hello 3 | Hello

Question 2:
What is the expected output of the following code?

```
def func(x):
    try:
        x = x / x
    except:
        print('a', end='')
    else:
        print('b', end='')
    finally:
        print('c', end='')

func(1)
func(0)
```

A. acac
B. bcac
C. bac
D. bca
E. bcbc

Question 3:
The function body is missing.
What snippet would you insert in the line indicated below:

```
def func(number):
    # insert your code here

print(func(7))
```

A. print ('number')
B. print (number)
C. return number
D. return 'number'

Question 4:
The digraph written as #! is used to:

A. tell an MS Windows OS how to execute the contents of a Python file.
B. create a docstring.
C. tell a Unix or Unix-like OS how to execute the contents of a Python file.
D. make a particular module entity a private one.

Question 5:
A data structure described as LIFO is actually a:

A. tree
B. heap
C. stack
D. list

Question 6:
What is the expected output of the following code?

```
x = '\''
print(len(x))
```

A. 0
B. The code is erroneous.
C. 2
D. 1

Question 7:
The ABC organics company needs a simple program that their call centre will use to enter survey data for a new coffee variety. The program must accept input and return the average rating based on a five-star scale. The output must be rounded to two decimal places. You need to complete the code to meet the requirements.

```
sum = count = done = 0
average = 0.0

while done != -1:
    rating = XXX
    if rating == -1:
        break
    sum += rating
    count += 1

average = float(sum / count)

YYY + ZZZ
```

What should you insert instead of XXX, YYY and ZZZ?

A.	1 | XXX -> input (‘Enter next rating (1-5), -1 for done’) 2 | YYY -> print (‘The average star rating for the new coffee is: ‘ 3 | ZZZ -> format (average. ‘.2d’))
‘B.	1 | XXX -> float (input (‘Enter next rating (1-5), -1 for done’)) 2 | YYY -> output (‘The average star rating for the new coffee is: ‘ 3 | ZZZ -> format (average, ‘.2d’))
C.	1 | XXX -> print (input (‘Enter next rating (1-5), -1 for done’)) 2 | YYY -> print (‘The average star rating for the new coffee is : ‘ 3 | ZZZ -> format (average, ‘.2f’))
D.	1 | XXX -> float (input (‘Enter next rating (1-5), -1 for done’)) 2 | YYY -> print (‘The average star rating for the new coffee is : ‘ 3 | ZZZ -> format (average, ‘.2d’))
E.	1 | XXX -> float (input (‘Enter next rating (1-5), -1 for done’)) 2 | YYY -> print (‘The average star rating for the new coffee is: ‘ 3 | ZZZ -> format (average, ‘.2f’))
F.	1 | XXX -> float (input (‘Enter next rating (1-5), -1 for done’)) 2 | YYY -> printline(‘The average star rating for the new coffee is: ‘ 3 | ZZZ -> format (average, ‘.2f’))

Question 8:
What is the expected output of the following code?

```
list1 = [1, 3]
list2 = list1
list1[0] = 4
print(list2)
```

A. [1, 4]
B. [4, 3]
C. [1, 3, 4]
D. [1, 3]

Question 9:
What is the expected output of the following code?

```
import math

result = math.e != math.pow(2, 4)
print(int(result))
```

A. False
B. 0
C. True
D. 1

Question 10:
What is the expected output of the following code?

```
data = ['Peter', 404, 3.03, 'Wellert', 33.3]
print(data[1:3])
```

A. [404, 3.03]
B. [‘Peter’ , 404,3.03, ‘Wellert’ , 33.3]
C. [‘Peter’ , ‘Wellert’]
D. None of the above.

Question 11:
What is the expected output of the following code?

```
print(list('hello'))
```

A. hello
B. None of the above.
C. [‘h’ ‘e’ ‘l’ ‘l’ ‘o’]
D. [‘h’ , ‘e’ , ‘l’ , ‘l’ , ‘o’]
E. [h, e, l, l, o]

Question 12:
What will be the output of the following code snippet?

```
print(3 / 5)
```

A. None of the above.
B. 6/10
C. 0.6
D. 0

Question 13:
What is the default return value for a function that does not explicitly return any value?

A. None
B. public
C. Null
D. int

Question 14:
What is the expected output of the following code?

```
x = True
y = False
z = False

if not x or y:
    print(1)
elif not x or not y and z:
    print(2)
elif not x or y or not y and x:
    print(3)
else:
    print(4)
```

A. 1
B. 4
C. 3
D. 2

Question 15:
What would you insert instead ??? so that the program checks for even numbers?

```
if ???:
    print('x is an even number')
```

A. x % 2 == 1
B. x % 2 == 0
C. x % 1 == 2
D. x % 'even' == True

Question 16:
What is the expected output of the following code?

```
nums = [3, 4, 5, 20, 5, 25, 1, 3]
nums.pop(1)
print(nums)
```

A. [3, 1, 25, 5, 20, 5, 4]
B. [3, 4, 5, 20, 5, 25, 1, 3]
C. [1, 3, 4, 5, 20, 5, 25]
D. [1, 3, 3, 4, 5, 5, 20, 25]
E. [3, 5, 20, 5, 25, 1, 3]

Question 17:
An operator able to check whether two values are not equal is coded as:

A. not ==
B. < >
C. !=
D. =/=

Question 18:
What is the expected output of the following code?

```
def func(p1, p2):
    p1 = 1
    p2[0] = 42

x = 3
y = [1, 2, 3]

func(x, y)

print(x, y[0])
```

A. 1 1
B. 3 42
C. 3 1
D. 1 42
E. The code is erroneous.

Question 19:
What will be the output of the following code snippet?

```
x = 2
y = 1
x *= y + 1
print(x)
```

A. 2
B. 4
C. None
D. 1
E. 3

Question 20:
What is the expected output of the following code if the user enters 2 and 4?

```
x = input()
y = input()
print(x + y)
```

A. 4
B. 2
C. 24
D. 6

Question 21:
What is the expected output of the following code?

```
x = [[[1, 2], [3, 4]], [[5, 6], [7, 8]]]

def func(data):
    res = data[0][0]
    for da in data:
        for d in da:
            if res < d:
                res = d
    return res

print(func(x[0]))
```

A. 4
B. The code is erroneous.
C. 2
D. 6
E. 8

Question 22:
What is the correct command to shuffle the following list?

```
import random
people = ['Peter', 'Paul', 'Mary', 'Jane']
```

A. random.shuffleList (people)
B. shuffle(people)
C. people.shuffle( )
D. random.shuffle (people)

Question 23:
Entering the try: block implies that:

A. some of the instructions from this block may not be executed.
B. none of the instructions from this block will be executed.
C. the block will be omitted.
D. all of the instructions from this block will be executed.

Question 24:
What will be the output of the following code snippet?

```
a = [1, 2, 3, 4, 5, 6, 7, 8, 9]
print(a[::2])
```

A. [8, 9]
B. [1, 2]
C. [1, 3, 5, 7, 9]
D. [1, 2, 3]

Question 25:
What is the expected output of the following code?

```
x = 1 / 2 + 3 // 3 + 4 ** 2
```

```
print(x)
```

A. 17
B. 8
C. 17.5
D. 8.5

Question 26:
What is the output of the following code snippet?

```
def test(x=1, y=2):
    x = x + y
    y += 1
    print(x, y)

test(2, 1)
```

A. The code is erroneous.
B. 2 3
C. 3 3
D. 1 3
E. 3 2

Question 27:
Which of the following variable names is illegal?

A. in
B. in_
C. In
D. IN

Question 28:
What is the expected output of the following code?

```
from datetime import date

date_1 = date(1992, 1, 16)
date_2 = date(1991, 2, 5)

print(date_1 - date_2)
```

A. 345, 0:00:00
B. 345 days, 0:00:00
C. 345 days
D. 345

Question 29:
What value will be assigned to the x variable?

```
z = 3
y = 7
x = y < z and z > y or y > z and z < y
```

A. 0
B. False
C. True
D. 1

Question 30:
What is the expected output of the following code?

```python
file = open('data.txt', 'w+')
print('Name of the file: ', file.name)

s = 'Peter Wellert\nHello everybody'
file.write(s)
file.seek(0)
for line in file:
    print(line)

file.close()
```

A.	1 | Peter Wellert Hello everybody
B.	The code is erroneous.
C.	1 | Name of the file: data.txt 2 | Peter Wellert 3 | 4 | Hello everybody
D.	1 | Peter Wellert 2 | Hello everybody

Question 31:
What is the expected output of the following code?

```
from datetime import datetime

datetime = datetime(2019, 11, 27, 11, 27, 22)
print(datetime.strftime('%y/%B/%d %H:%M:%S'))
```

A. 2019/11/27 11:27:22
B. 19/November/27 11:27:22
C. 2019/Nov/27 11:27:22
D. 19/11/27 11:27:22

Question 32:
What will be the output of the following code snippet?

```
d = {}
d[1] = 1
d['1'] = 2
d[1] += 1

sum = 0

for k in d:
    sum += d[k]

print(sum)
```

A. 4
B. 1
C. 3
D. 2

Question 33:
What is the expected output of the following code?

```
num = '7' * '7'
print(num)
```

A. The code is erroneous.
B. 7777777
C. 49
D. 77

Question 34:
What is the expected output of the following code?

```
x = [0, 1, 2]
x.insert(0, 1)
del x[1]
print(sum(x))
```

A. 3
B. 4
C. 5
D. 2

Question 35:
The Exception class contains a property named args and it is a:

A. list
B. tuple
C. dictionary
D. string

Question 36:
The 0o prefix means that the number after it is denoted as:

A. octal
B. decimal
C. binary
D. hexadecimal

Question 37:
What is the expected output of the following code?

```
x = 1 // 5 + 1 / 5
```

```
print(x)
```

A. 0.4
B. 0.2
C. 0.0
D. 0

Question 38:
Which of the following for loops would output the below number pattern?

```
11111
22222
33333
44444
55555
```

A.	1 | for i in range(1, 6): 2 | print(i, i, i, i, i)
B.	1 | for i in range(1, 5) 2 | print(str(i) * 5
C.	1 | for i in range(1, 6): 2 | print(str(i) * 5)
D.	1 | for i in range(0, 5): 2 | print(str(i) * 5)

Question 39:
Which of the following commands can be used to read n characters from a file?

A. file .read(n)
B. n = file.readline( )
C. n = file.read( )
D. file. readline(n)

Question 40:
isalnum() checks if a string contains only letters and digits, and this is:

A. A method
B. A function
C. A module
D. None of the above.

Question 41:
You want to write a code snippet to read the total data from a text file and print it to the monitor. What snippet would you insert in the line indicated below:

```
1  try:
2      file = open('data.txt', 'r')
3      # insert your code here
4      print(data)
5  except:
6      print('Something went wrong!')
```

A. data = file.read( )
B. data = file.load( )
C. data = file.readline( )
D. data = file.readlines( )

Question 42:
Knowing that a function named randint() resides in the module named random choose the proper way to import it:

A. import randint
B. import randint from random
C. from randint import random
D. from random import randint

Question 43:
What is the expected output of the following code?

```
x = 1
y = 2
x, y, z = x, x, y
z, y, z = x, y, z
print(x, y, z)
```

A. 1 2 1
B. 1 2 2
C. 1 1 2
D. 2 1 2

Question 44:
Assuming that the open() invocation has gone successfully, the following snippet will:

```
for x in open('file', 'rt'):
    print(x)
```

A. cause an exception.
B. read the file line by line.
C. read the whole file at once.
D. read the file character by character.

Question 45:
What is the expected output of the following code?

```
def func(num):
    res = '*'
    for _ in range(num):
        res += res
    return res


for x in func(2):
    print(x, end='')
```

A. The code is erroneous.
B. *
C. **

D. ****

Question 46:
How many stars will the following code print to the monitor?

```
i = 0
while i <= 3:
    i += 2
    print('*')
```

A. zero
B. one
C. three
D. two

Question 47:
What is the expected output of the following code?

```
def func():
    try:
        print(23)
    finally:
        print(42)

func()
```

A.	1 | 42 2 | 23
B.	23
C.	42
D.	1 | 23 2 | 42
E.	None of the above.

Question 48:
The value thirty point eleven times ten raised to the power of nine should be written as:

A. 30E11.9
B. 30.11*10^9
C. 30.11E9.0
D. 30.11E9

Question 49:

What is the expected output of the following code?

```
def func(x):
    return 1 if x % 2 != 0 else 2

print(func(func(1)))
```

A. None
B. The code is erroneous.
C. 1
D. 2

Question 50:
When a module is imported, its contents:

A. are ignored.
B. are executed once.
C. are executed depending on the contents.
D. are executed as many times as they

Question 51:
What is the expected output of the following code?

```
z = y = x = 1
print(x, y, z, sep='*')
```

A. x y z
B. 1 1 1
C. The code is erroneous.
D. x*y*z
E. 111*
F. 1*1*1

Question 52:
How many stars will the following snippet print to the monitor?

```
i = 4
while i > 0:
    print('*')
    if i == 2:
        break
else:

    print('*')
```

A. The snippet will enter an infinite loop.
B. 0
C. 2
D. 1

Question 53:
What is the expected output of the following code?

```python
num = 1

def func():
    num = num + 3
    print(num)

func()

print(num)
```

A. 4 1
B. 1 1
C. The code is erroneous.
D. 4 4
E. 1 4

Question 54:
You develop a Python application for your company. You have the following code.

```python
def main(a, b, c, d):
    value = a + b * c - d
    return value
```

Which of the following expressions is equivalent to the expression in the function?

A. a + ((b* c) - d)
B. (a + (b * c)) - d
C. (a + b) * (c - d)
D. None of the above.

Question 55:
Consider the following code snippet:

```
w = bool(23)
x = bool('')
y = bool(' ')
z = bool([False])
```

Which of the variables will contain False?

A. z
B. y
C. x
D. w

Question 56:
The part of your code where you think an exception may occur should be placed inside:

A. the except: branch
B. the exception: branch
C. the try: branch
D. None of the above.

Question 57:
What will be the output of the following code snippet?

```
x = 1
y = 2
z = x
x = y
y = z
print(x, y)
```

A. 1 2
B. 2 1
C. 1 1
D. 2 2